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Mental arithmetic skill development in primary school: The importance of number processing abilities and general cognitive abilities

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ABSTRACT

The purpose of this study was to pinpoint which mixture of cognitive abilities and number abilities underlies young children's early mental arithmetic learning (i.e., skill development) and to examine to what extent this mixture is akin to the mixture underlying children's early arithmetic performance. A total of 265 children were assessed on counting knowledge, symbolic magnitude comparison, number line estimation, logical reasoning, verbal working memory, spatial processing, phonological processing, and general processing speed. One year later in first grade, the children's mental arithmetic ability was assessed, and it was then reassessed in second grade. A latent change score model showed that arithmetic performance was supported by counting knowledge, number line estimation, logical reasoning, spatial processing, phonological processing, and general processing speed, whereas arithmetic development was only supported by verbal working memory. These results demonstrate that the mixture of abilities underlying arithmetic development and arithmetic performance are rather different. Mental arithmetic performance in Grade 1 is equally dependent on a combination of both number abilities and cognitive abilities, whereas mental arithmetic development between first grade and second grade is only supported by one cognitive ability, verbal working memory.

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Introduction

It has long been established that children's numerical and arithmetic skills vary greatly already at school entry and throughout all school years (Anders et al., 2013; Duncan et al., 2007). In view of this, a bulk of research has focused on accounting for these individual differences by identifying predictors of children's mathematical performance (e.g., Fuchs et al., 2010a; Geary et al., 2017; Träff et al., 2020). This research has advanced our understanding by providing converging evidence that the resources underlying children's mathematical performance consists of a mixture of general cognitive abilities and number abilities (Fuchs et al., 2010a; Östergren & Träff, 2013). Contrary to the large amount of cross-sectional evidence, few longitudinal developmental studies have examined the cognitive resources supporting children's mathematical skill learning. Against this background, the purpose of the study was to expand our theoretical understanding of children's development of early arithmetic skills by pinpointing which mixture of abilities underlies young children's early arithmetic learning (i.e., development) and to examine to what extent this mixture is akin to the mixture underlying children's early arithmetic performance.

The current study was founded on the assumption that arithmetic *performance* and arithmetic *learning* (*development*) involve different processes to some extent. When young children solve arithmetic problems (i.e., arithmetic performance), they use their existing limited arithmetic knowledge and skills (e.g., strategies, ways of thinking) that they have acquired so far. However, arithmetic learning or development involves increasing children's existing knowledge and skills in order to solve arithmetic problems more accurately and efficiently by acquiring, for example, more effective strategies and ways of thinking as well as executing them in an increasingly effective way (Siegler, 2000). The notion that performance, and learning involve different processes and thereby may draw on different constellations of abilities is supported by prior longitudinal studies showing that children with the same initial math performance at the first time point can display vastly different patterns of development (i.e., learning) at future time points (Gesuelli & Jordan, 2024; Morgan et al., 2009; Resnick et al., 2016; Salaschek et al., 2014). These findings demonstrate that the rate of learning is not entirely dependent on initial performance level. Thus, to study individual differences in arithmetic learning or development, it is necessary to perform longitudinal studies and to analyze individual differences in skill improvement (i.e., learning) across measurement points.

The purpose of this study was addressed by testing two different accounts: The number foundations account proposed by Fuchs et al. (2010b; see also von Aster & Shalev, 2007) claims that children's early stages of basic arithmetic learning is, akin to early arithmetic performance, uniquely supported by both domain-specific number abilities and domain-general cognitive abilities. However, early arithmetic learning (and performance) is, contrary to later and more advanced arithmetic learning, relatively more dependent on number abilities than general cognitive abilities because it is more firmly connected to the core number magnitude system than later and more advanced arithmetic learning.

The mixed foundations account proposed by Geary et al. (2017) and Lee and Bull (2016) states that, akin to the number foundations account (Fuchs et al., 2010b), young children's early stage of arithmetic learning and performance is uniquely supported by both domain-specific number abilities and domain-general cognitive abilities. However, due to young children's low mathematical knowledge levels, their early stage of arithmetic performance and learning is equally reliant on number abilities and general cognitive abilities (cf. Ackerman, 1988; Siegler, 2000). Thus, the number foundations account and the mixed foundations account differ on their view on the relative importance of domain-general abilities and domain-specific abilities in relation to early arithmetic learning and performance.

One reason for studying arithmetic learning in young children is that it constitutes an important aspect of cognitive development. Studying learning therefore not only will expand our understanding of mathematical development per se but also may provide knowledge about children's cognitive development in general (Munakata & Johnson, 2006; Piaget, 1952). That is, findings concerning which mixture of cognitive abilities underlies early mental arithmetic learning (development) may be applicable to understand young children's development within other academic domains as well as cognitive domains (cf. Ackerman, 1988). In other words, what is the primary force of development

in children: domain-specific or domain-general abilities or both? Support for the number foundations account would suggest that domain-specific abilities are the primary determinant of development, whereas support for the mixed foundations account would suggest that development is equally determined by domain-specific and domain-general abilities.

Apart from this academic interest, knowledge of which cognitive resources are important for *developing* a higher level of proficiency in early arithmetic versus *performing* at a high level of proficiency in early arithmetic may also be important to detect young children at risk of developing mathematical difficulties and inform how to design and apply targeted intervention to help these children (Gersten et al., 2005; Jordan et al., 2009).

To accomplish this study, a mental arithmetic measure and a broad set of predictors were employed. A total of 265 preschool children (aged 6 years) were assessed on number abilities and cognitive abilities. Children's arithmetic skills were assessed 1 year later in first grade, and then they were reassessed in second grade. The two measurement points were used to calculate latent change scores as an index of skill development.

Early arithmetic development and performance in young children: The relative unique support of number abilities and cognitive abilities

Early arithmetic performance

Prior cross-sectional studies demonstrate that the underlying mechanism of young children's early arithmetic performance consists of a combination of both general cognitive abilities and number abilities (Geary et al., 2017; Jordan et al., 2010; Martin et al., 2014; Östergren & Träff, 2013). These findings align with the claim shared by the two accounts (i.e., Fuchs et al., 2010b; Geary et al., 2017). Prior research also indicates that 6-year-olds' written multi-digit calculation performance is uniquely supported by both types of abilities (Jordan et al., 2010; Martin et al., 2014). However, existing cross-sectional research focusing on the relative unique support of number abilities and cognitive abilities has generated contradictory findings in relation to the two accounts. For example, consistent with the number foundations account (Fuchs et al., 2010b), Fuchs et al. (2010a) found that single-digit arithmetic fluency was predicted by two number abilities (number sets test: 2.9%; number line estimation: 37.2%) but none of the general cognitive abilities included in the studies. Furthermore, Martin et al. (2014) observed that number abilities (digit comparison: 8.7%; symbolic number naming: 11.2%) accounted for 19.9% of the variation in 6-year-olds' arithmetic performance, whereas verbal working memory only accounted for 1.8% unique variance. In contrast to research supporting the number foundations account, Östergren and Träff (2013) and Geary et al. (2017) found that 6-year-olds' and second graders' arithmetic skills, respectively, were relatively more reliant on cognitive abilities (e.g., working memory, intelligence) than number abilities. Moreover, Jordan et al. (2010) observed that first graders' written arithmetic calculation is equally supported by number abilities and cognitive abilities (see also Lee & Bull, 2016), which is consistent with the mixed foundations account (Geary et al., 2017).

Early arithmetic development

In addition to cross-sectional research, a few researchers have attempted to map the mixture of resources underlying children's mathematical development (i.e., learning). One seminal study was performed by Fuchs and colleagues (2010b), where they assessed development of double-digit mental calculation between the fall and spring semesters of first grade. A latent change score was computed as measures of arithmetic development and analyzed with multiple regression analysis. Development (R^2 = .33) was predominantly supported by number abilities (number sets test: 8.4%;number line estimation: 4.8%) and slightly supported by verbal working memory (2.0%). Based on these findings, Fuchs et al. (2010b) developed their number foundations account presented above.

Support for the number foundations account has been provided by Zhang et al. (2014), who, akin to Fuchs et al. (2010b), used specific outcome measures to account for variation in basic arithmetic development (e.g., 2 + 1, 3 + 4 + 6, 4 − 1, 20 – 2 − 4). Zhang et al. (2014) followed children's arithmetic development from first grade to third grade. Unfortunately, their set of predictors (counting knowledge, spatial processing, phonemic awareness, receptive vocabulary, and letter knowledge) was less

extensive and quite different from the set used by Fuchs et al. (2010b) and prior research focusing on performance. Despite this, growth curve modeling indicated that variation in arithmetic development (R^2 = .23) was mainly supported by counting knowledge (19.36%) and, to a lesser extent, by spatial processing ability (3.24%), whereas counting knowledge (15.21%), spatial processing ability (3.61%), and letter knowledge (2.25%) supported arithmetic performance (R^2 = .37). Thus, Zhang et al.'s (2014) findings provided strong support for the number foundations account (Fuchs et al., 2010b) given that early arithmetic development and performance are supported by both domain-specific number abilities and domain-general cognitive abilities, but with a relatively greater dependence on number abilities rather than general cognitive abilities.

Additional evidence for the number foundations account has been presented by Aunola et al. (2004), who examined skill development in children from preschool to second grade using a general measure of mathematics. However, their test battery only included measures of counting knowledge skills, listening comprehension, metacognitive knowledge, and visual attention. Latent growth modeling showed that skill development (R^2 = .23) relied on counting knowledge (19.36%) and visual selective attention (2.25%), whereas counting knowledge (37.4%), listening comprehension (2.6%), and metacognitive knowledge (2.9%) supported performance (R^2 = .58).

Geary (2011) obtained support for the mixed foundations account (Geary et al., 2017) in a study employing the numerical operations test to follow children's development from kindergarten to Grade 5. Multilevel modeling revealed that performance was aided by two number abilities (number sets test: 22.9%; pseudo-counting: 3.61%), two arithmetic skills (complex addition procedure: 13.0% and addition decomposition: 44.89%), and four cognitive abilities (verbal working memory: 8.41%; processing speed: 6.76%; visuospatial short-term memory: 6.76%; intelligence: 5.76%). Development relied on verbal working memory (11.6%), number line estimation (1.7%), addition fact retrieval (14.4%), and addition decomposition strategy (4.8%).

Research by Xenidou-Dervou et al. (2018) and Träff et al. (2023) have displayed mixed results in relation to the two accounts. Xenidou-Dervou et al. (2018) employed a general mathematics test to track skills development over four time points from Grade 1 to Grade 2. Consistent with the mixed foundations account (Geary et al., 2017), latent growth modeling revealed that four cognitive abilities (verbal working memory: 1.44%; verbal short-term memory: 1.69%; visuospatial short-term memory: 1.44%; logical reasoning: 2.25%) and three number abilities (counting knowledge skills: 5.29%; nonsymbolic number processing: 1.44%; symbolic number processing: 1.69%) provided equal support to initial performance (R^2 = .52). However, only symbolic approximate addition (6.8%) supported mathematics development (R^2 = .11), which contradicts both accounts given that they claim that basic arithmetic learning and performance are uniquely supported by both domain-specific number abilities and domain-general cognitive abilities. Similar findings were obtained by Träff et al. (2023) when they attempted to identify key cognitive abilities contributing to 6-year-old children's early single-digit arithmetic development from preschool to first grade. Contrary to both accounts, a latent change score model displayed that arithmetic development (R^2 = .252) was only supported by spatial processing (19.6%). Consistent with the number foundations account (Fuchs et al., 2010b), arithmetic performance (R^2 = .67) was supported by one number ability (counting knowledge: 25.3%) and three general cognitive abilities (verbal logical reasoning: 4.6%; nonverbal logical reasoning: 2.8%; spatial processing: 2.4%).

In summary, evidence aligns with the notion, shared by the two accounts, that the resources underlying young children's early arithmetic performance and development consists of a mixture of cognitive abilities and number abilities. However, consistent with the number foundations account, most developmental evidence shows that early mathematical development is relatively more dependent on number abilities than cognitive abilities (Aunola et al., 2004; Fuchs et al., 2010b; Geary, 2011; Zhang et al., 2014). In a similar manner, most prior studies indicate that early arithmetic performance is relatively more reliant on number abilities than cognitive abilities. Consistent with the mixed foundations account, a few studies indicate that cognitive abilities are more important than number abilities, whereas a few other studies indicate that the two types of abilities are equally important (see Jordan et al., 2010; Xenidou-Dervou et al., 2018).

Counting knowledge, working memory, logical reasoning (IQ), language skills (phonological processing), and spatial processing have, with some consistency, emerged as key abilities underlying early

arithmetic performance. However, the few existing developmental studies do not display a stable set of abilities that have been found to consistently support children's early mathematical development. This is partially due to some researchers having used specific arithmetic measures akin to Fuchs et al. (2010b), whereas other researchers have used broad measures of mathematics and/or quite different constellations of predictor abilities. Thus, further research is required to obtain a more comprehensive understanding of the cognitive resources underlying children's mathematical development and learning.

The specific roles of number abilities and cognitive abilities during early arithmetic learning

Domain-specific number abilities

Von Aster and Shalev's (2007) four-step developmental model of numerical cognition (see also Dehaene, 1992) provides some indications as to what roles the different number abilities play during mathematical learning. According to the model, humans are born with a core system of cardinal magnitude (Step 1), enabling the performance of preverbal magnitude functions (e.g., subitizing, approximation, magnitude comparison). The core system is foundational for developing symbolic number magnitude systems (cf. Dehaene, 1992). Symbolic number systems develop through the learning of number words and Arabic numerals and by integrating them with the core magnitude system, resulting in a verbal number system (Step 2) and an Arabic number system (Step 3). These two symbolic number systems represent children's cardinal understanding of number symbols and is foundational for developing a symbolic mental number line (Step 4). This spatially oriented number line represents, besides cardinality, an understanding of ordinality (i.e., what position a number has in a sequence of numbers) as a second core principle of numbers. One main assumption of the model is that the learning of number symbols (counting words and Arabic numerals) and integrating them with the core magnitude system in Steps 2 to 4 is increasingly reliant on support from general cognitive abilities such as language, executive functions, and working memory. The symbolic number systems are, in turn, the basis for acquiring and performing basic number processing skills and mathematical skills.

Counting knowledge. During the initial stages of arithmetic performance and learning, children employ finger and verbal counting strategies to solve arithmetic problems (Geary & Hoard, 2005; Siegler & Shrager, 1984). This notion relates to the verbal number system of von Aster and Shalev's (2007) model responsible for rote counting, enumeration and solving basic arithmetic problems using counting strategies or direct retrieval. Consistent with this notion, evidence demonstrated that counting skill is involved in children's early arithmetic performance and development (Aunola et al., 2004; Geary, 2011; Zhang et al., 2014; see Xenidou-Dervou et al., 2018, for contradictory findings).

Symbolic magnitude comparison. In learning to solve written arithmetic problems (e.g., 2 + 3, 4 − 2), one must be able to identify and access the underlying magnitudes of the Arabic numerals. Consistent with this assumption, related to Step 3 of von Aster and Shalev's (2007) model, studies show that symbolic magnitude comparison accounts for unique variation in early arithmetic performance (Martin et al., 2014; Xenidou-Dervou et al., 2018). However, it is still unclear to what extent rapid and accurate access to the underlying magnitudes of Arabic numerals contributes to arithmetic learning (development) given that only one developmental study has included it in the test battery (Xenidou-Dervou et al., 2018).

Number line estimation. Number line estimation is a complex number processing task assumed to tax the symbolic mental number line (Step 4) of von Aster and Shalev's (2007) model. Thus, the number line estimation task assesses both cardinality (i.e., magnitude) and an ordinal understanding of numbers, and therefore it may play several roles in early arithmetic learning. A well-established ordinal symbolic mental number line is important given that the child is assumed to move mentally along it during arithmetic thinking (von Aster & Shalev, 2007). During arithmetic learning, the quality of the mental number line should be especially critical as children strive to expand their conceptual and procedural knowledge by acquiring more effective solution strategies and executing them in an increasingly effective way (Siegler, 2000). In line with this notion, prior research shows that number

line estimation supports early arithmetic performance and development in children beyond the support of cardinality and general cognitive processing (Fuchs et al., 2010a, 2010b; Geary, 2011).

Domain-general cognitive abilities

The number foundations account (Fuchs et al., 2010b) and the mixed foundations account (Geary et al., 2017) emphasize that children's arithmetic performance and learning is supported by three cognitive abilities: language, executive functions, and (especially) working memory. Prior research also emphasizes these three abilities and proposes several other abilities to form the foundation of children's arithmetic learning (e.g., processing speed, logical reasoning, spatial processing; Fuchs et al., 2010b; Geary, 2011; Geary et al., 2017; Träff et al., 2023).

Working memory. The working memory construct implicates the simultaneous maintenance and processing of relevant information, inhibiting activation of irrelevant information, and shifting between operations or strategies during the performance of complex cognitive tasks (Baddeley, 1997; Engle et al., 1999; Shah & Miyake, 1996). In view of this construct's multiple functions, it should play several roles during arithmetic performance and learning. A straightforward role entails the temporary storage of relevant numerical information and support for finger counting and verbal counting strategies (Geary, 2013). Working memory may also facilitate children's ability to learn new effective strategies and select the most effective strategy given the task at hand and to shift between strategies, as well as execute them in increasingly effective ways. Learning new arithmetic concepts and new effective strategies should require support from a flexible and efficient mental workspace where new knowledge can be integrated with existing knowledge (Gathercole & Alloway, 2004; Lee & Bull, 2016). Consistent with these roles, research demonstrates that working memory tasks provide a unique contribution to early arithmetic performance (e.g., Fuchs et al., 2010a; Geary, 2011; Martin et al., 2014; Xenidou-Dervou et al., 2018). However, only a few studies have found that working memory uniquely supports arithmetic development (e.g., Fuchs et al., 2010b; Geary, 2011).

Language. Compelling evidence demonstrates a link between young children's language processes (e.g., comprehension, vocabulary) and mathematics performance and development (Aunola et al., 2004; Fuchs et al., 2010b; Jordan et al., 2010). In addition, several studies have used phonological processing tasks as a proxy of language ability and observed that phonological processing supports children's arithmetic performance (e.g., De Smedt et al., 2010; Hecht et al., 2001). Still, no study has yet detected a link between phonological processing and arithmetic learning. It has been hypothesized that the phonological processing–arithmetic link may reflect that more distinct phonological representations enable children to solve arithmetic problems more efficiently by providing distinct and accurate verbal number representations as well as accurate long-term storage of arithmetic facts (De Smedt et al., 2010; see also Dehaene, 1992).

Logical reasoning. It stands to reason that logical reasoning should play a vital role during the process of acquiring sufficient arithmetic knowledge and skills, because children must understand the underlying logic of adding and subtracting (Geary, 2013). However, learning arithmetic and its logical basis requires that children first obtain a conceptual understanding of several important concepts (i.e., cardinality, ordinality, place value, the base-10 number system, arithmetic operators, and the equal sign) and how different arithmetic operations are related (Baroody & Dowker, 2003). All this conceptual learning and new strategies and ways of thinking are assumed to be prerequisites for arithmetic learning and thereby put strain on children's logical reasoning ability (Geary, 2013; Geary et al., 2017; Lee & Bull, 2016). This rationale is supported by a large body of cross-sectional evidence obtained from different age groups (5.5–12 years; cf. Geary, 2011; Jordan et al., 2010; Xenidou-Dervou et al., 2018). However, Geary (2011) and Xenidou-Dervou et al. (2018) failed to detect a link between children's logical reasoning ability and mathematical development. Thus, despite the strong rationale, the empirical link between young children's logical reasoning and early arithmetic learning is still unclear.

General processing speed. It is generally agreed that general processing speed, in addition to working memory and intelligence (logical reasoning), is a key ability involved in nearly all learning that

requires rapid identification, activation, and access to information or other forms of rapid information processing (Carroll, 1993; Geary, 2011; Gottfredson, 1997). Prior research provides some evidence that general processing speed is uniquely supportive of arithmetic performance in children (Bull & Johnston, 1997; Geary, 2011). Still, there is currently no evidence indicating that general processing speed is supportive of arithmetic learning (development). In theory, however, it should enable rapid execution of new effective strategies and ways of thinking during development of more efficient arithmetic skills (Siegler, 2000).

Spatial processing. One account of how spatial processing supports early arithmetic learning is that young children may convert simple arithmetic problems into spatial representations as a solution strategy. The representations are then matched against the spatially oriented mental number line (Step 4; von Aster & Shalev, 2007) to enable effective and concurrent cardinal and ordinal processing (see Barrouillet et al., 2008; Mathieu et al., 2016). It is also possible that this matching process involves using the mental number line as a monitoring mechanism for determining whether an answer is realistic or not (Hegarty & Kozhevnikov, 1999; Marghetis et al., 2014; Xenidou-Dervou et al., 2018). The spatial processing–mathematics link is supported by several studies. For example, Gunderson et al. (2012) and Lachance and Mazzocco (2006) found that children's mathematical performance is supported by spatial processing (mental rotation) ability from kindergarten to Grade 12. More important, Zhang et al. (2014) and Träff et al. (2023) observed that young children's early arithmetic performance and development is supported by spatial processing ability.

The current study

This study sought to expand our knowledge concerning the mixture of cognitive abilities and number abilities underlying children's early arithmetic performance and learning by testing and comparing the number foundations account and the mixed foundations account. This was accomplished by employing a broad set of number and cognitive abilities based on findings from prior developmental studies (i.e., Aunola et al., 2004; Fuchs et al., 2010b; Geary, 2011; Zhang et al., 2014). In addition, because developing basic single (2 + 3), and double-digit (15 + 19) arithmetic skills is the primary mathematical learning task during Grade 1, an age-appropriate mental arithmetic measure and test procedure was designed to obtain reliable and valid scores of mental arithmetic. This allowed us to capture the very large variation in arithmetic ability that exists among first and second graders (Anders et., 2013; Duncan et al., 2007). This was achieved by presenting all arithmetic problems, which became successively more difficult, on sheets of paper. Each problem was also read out to children, who needed to provide an oral response. Testing was untimed but was stopped after three consecutive errors. This test procedure was assumed to eliminate unnecessary requirements (e.g., remembering the operands in the problem, providing written responses, general time pressure) that are not directly related to arithmetic ability but that many young children may find demanding and thereby may hamper their performance.

In alignment with both the number foundations account and the mixed foundations account (Hypothesis 1), children's early mental arithmetic learning (i.e., development) and mental arithmetic performance should rely on a combination of domain-specific number abilities and domain-general cognitive abilities. However, according to the number foundations account (Hypothesis 2a), early mental arithmetic learning and performance should be relatively more dependent on number abilities than general cognitive abilities, whereas the mixed foundations account (Hypothesis 2b) claims that early arithmetic learning and performance should be equally reliant on number abilities and general cognitive abilities.

The study also addressed this question: To what extent is the mixture of abilities underlying development akin to the mixture underlying children's early mental arithmetic performance?

The current study has important theoretical implications. If Hypothesis 1 and Hypothesis 2a or 2b are supported—that is, arithmetic development and performance are supported by the same or similar combinations of abilities—strong support should be provided to contemporary frameworks of numerical and mathematical development (Fuchs et al., 2010b; Geary et al., 2017; LeFevre et al., 2010). If, on the other hand, arithmetic development is supported by a completely different mixture of abilities

compared with arithmetic performance, or even worse is not supported by a mixture of number abilities and cognitive abilities, the models would be contradicted. This would imply that new developmental models need to be developed.

Method

Participants

A total of 265 children (137 boys and 128 girls) from 24 public schools in a town located in the southern part of Sweden participated in this longitudinal study. During preschool (Measurement Point 1), the mean age was 6 years 9 months (*SD* = 3 months; range = 5 years 10 months to 6 years 10 months). In Grade 2 (Measurement Point 3), the mean age was 8 years 9 months (*SD* = 4 months; range = 8 years 0 months to 9 years 5 months). Children had normal hearing, had normal or corrected-to-normal visual acuity, and were fluent Swedish speakers. Informed consent to participate was obtained from the children's caregivers via a letter of consent that children brought home from school.

The research project was approved by the regional ethics committee in Linköping, Sweden.

General procedure

At 6 years of age (preschool), children were assessed on three number processing abilities (counting knowledge, symbolic number comparison, and number line estimation) and five general cognitive abilities (nonverbal logical reasoning, verbal working memory, mental rotation, general processing, and phonological processing). The tests were performed in the following order with all children: counting knowledge, nonverbal logical reasoning, symbolic number comparison, number line estimation, verbal working memory, general processing speed, phonological processing, and mental rotation. One year later in Grade 1, the children's mental arithmetic ability was assessed, and then it was reassessed in Grade 2.

All testing was performed at school in three or four individual test sessions that lasted approximately 25 to 35 min and with breaks that varied from 15 to 60 min (i.e., lunch break) to minimize fatigue effects. All testing was performed from January to June across all three measurement points. All instructions were presented orally. All testing was performed by 14 experimenters under the supervision of the second author.

Domain-specific predictors

Counting knowledge. Children's counting knowledge skills were assessed through six separate verbal tasks (cf. Östergren & Träff, 2013). Three forward counting sequence tasks were administered, requiring participants to (a) count from 1 to 50, (b) twice recite the counting sequence from 0 to 30, and (c) recite the counting sequence from specific lower to higher numbers (i.e., 8–13, 24–29, 63–68, and 85–90). A backward counting sequence task was also administered, (d) requiring participants to recite the reversed-order count list from three numbers (i.e., 10, 15, and 23) to 0. Two numerical order knowledge tasks were administered, requiring participants to (e) name the number immediately following the given cue (6, 15, 53, 69, or 99) and to name the number immediately prior to the given cue (9, 17, 28, 40, or 80). The dependent measure was the number of correctly solved items (a total of 20 for the six tasks), each awarded with 1 point. Split-half reliability was established as r_{sh} = .91.

Symbolic magnitude comparison. Children's symbolic magnitude comparison ability was assessed with a paper-and-pencil test (cf. Clarke et al., 2015; Clarke & Shinn, 2004; Sepúlveda et al., 2019). The task was to discriminate, as quickly and accurately as possible, a total of 28 Arabic digit pairs and to select the numerically larger digit by crossing it out. Two worksheets were administered, featuring near-distance (i.e., 1) digit pair trials (e.g., 7–8, 5–4) and far-distance (i.e., 4–5) trials (e.g., 8–3, 2–7). Because the error rate was low (*M* = 4.38%) and no accuracy–speed trade-off effect was observed ($r = -.26$), response time was used as the outcome measure instead of combining accuracy and

response time. The response times for both sheets of paper were combined into a single dependent measure. Split-half reliability was measured as r_{sh} = .90.

Number line estimation. A paper-and-pencil test was used to assess number line estimation acuity (cf. Siegler & Booth, 2004). Children were instructed to mark the location of a reference number on a 21-cm horizontal number line flanked by a start point (i.e., 0) and an end point (i.e., 100) at the ends. A total of 16 experimental problems (3, 64, 15, 72, 25, 6, 84, 33, 4, 48, 90, 23, 52, 8, 29, 57) selected from Siegler and Booth (2004) were administered on separate pages in a booklet. The reference number was presented 2 cm above the number line's midpoint. The dependent measure was the absolute error percentage, that is, the estimation accuracy in relation to a perfect linear function. Cronbach's alpha was established as.85.

Domain-general predictors

Nonverbal logical reasoning. A total of 35 items from the matrix reasoning subtest of Wechsler's Intelligence Scale for Children-IV (Wechsler, 2003) was used to assess participants' nonverbal logical reasoning ability. The items consisted of a visuospatial design matrix with one missing piece, to be identified either by pointing to it or by orally reporting one of five response alternatives. Testing was stopped if participants gave four consecutive incorrect answers or four incorrect answers over the course of 5 trials. The dependent measure was the number of correctly performed trials, for a maximum of 35 points. Split-half reliability was measured as r_{sh} = .87.

Verbal working memory. A complex verbal working memory span task (cf. Träff et al., 2023), featuring a word decision subtask, was administered. Participants listened as the experimenter read a list of words. Following each word, participants were instructed to indicate whether the word corresponded to an animal (e.g., answering "yes" for *cat* and "no" for *pencil*) and to remember the presented word. When all words had been presented and evaluated within the current span size, children were asked to orally repeat the presented words in the correct serial order. Two trials with different words were administered for each span size, such that a span size of two would entail 2 trials with two words each to evaluate (i.e., decide whether it represented an animal) and recall. The span size was incrementally increased (from the initial two) by one word if children had correctly recalled the words in at least 1 trial. Testing was stopped when children failed to recall the words of either trial in the correct serial order. Half of all words were animal target words. The dependent variable was the total number of words recalled in the correct serial order. Split-half reliability was measured as r_{sh} = .75.

Phonological processing. In this task (Taube et al., 1984), children were instructed to determine which part of a word had been eliminated from an original Swedish word (e.g., what has been removed from the word *elefant* [elephant] if only *fant* remains?). The task consisted of 15 words, but testing stopped after 4 consecutive failed trials. The number of correctly solved problems was the dependent measure, and the maximum score was 15. Split-half reliability was established as r_{sh} = .78.

Spatial processing. Children's spatial processing abilities were assessed with a mental rotation task (Träff et al., 2023). A total of 16 trials were administered, each consisting of five horizontally ordered alphabetical letters. The leftmost letter was the target, around which the following four rotated comparison items were compared (in one of six picture–plane angles: at 45°, 90°, 135°, 225°, 270°, and 315°). Half (i.e., 2) of the comparison items were correctly rotated variants of the target, whereas the other half of items were (incorrectly) visually mirrored representations. Using a pencil, children were asked to mark the two correct stimuli that corresponded to rotations of the target stimulus. They were allotted 2 min to complete as many trials as possible. The dependent variable was the number of correctly solved trials, where 1 point was awarded if both correct comparison stimuli were marked (for a total of 16 points). Split-half reliability was measured as r_{sh} = .92.

General processing speed. A paper-and-pencil number-matching task was employed to assess children's processing speeds (cf. Bull & Johnston, 1997). Children were provided with a sheet of paper with 15 rows of seven digits in each row. Children were asked to find the two identical digits in each row

and to draw a line across the two digits as fast as possible during a maximum period of 90 s. A processing speed index was calculated by dividing total response time (in seconds) by the number of correctly completed rows (i.e., mean seconds / row). It was not possible to compute any reliability index for this task because it was performed only once, and the total response time was used as the dependent measure.

Mental arithmetic

Children were asked to solve mental addition and subtraction problems presented horizontally on two sheets of paper: one for each operation (cf. Fuchs et al., 2010b). The addition problems (2 + 1, 3 + 2, 4 + 3, 2 + 4, 3 + 5, 6 + 4, 5 + 6, 7 + 8, 9 + 10, 11 + 12, 14 + 13, 16 + 14, 15 + 17, 19 + 17, 23 + 18, 36 + 15, 43 + 18, 64 + 32, 86 + 24, 92 + 16, 123 + 72, 141 + 93, 163 + 126, 247 + 356, 316 + 476, 436 + 673, 3236 + 8685, 5389 + 7279, and 13965 + 59376) and subtraction problems (3 – 1, 7 – 3, 5 – 2, 6 – 4, 9 – 5, 10 – 6, 11 – 5, 13 – 7, 15 – 7, 18 – 9, 16 – 8, 19 – 6, 17 – 9, 22 – 11, 32 – 14, 42 – 13, 58 – 21, 71 – 12, 85 – 17, 98 – 19, 112 – 36,137 – 49, ~~231~~ – ~~143,~~ 456 – 378, 683 – 495, 823 – 579, 2368 – 1479, 4273 – 2559, and 51463 – 32794) became successively more difficult.

Each child sat next to the experimenter who read the problems to the child (e.g., "three plus one?"; "four minus two?") while the child followed along on the paper. Testing always started with the addition problems, and children were instructed to provide an oral response (without the use of paper and pencil), that is, a number word. Children were free to solve the problems any way they wanted and had no time limit for their answers. The maximum score was 29 for both operations. Testing stopped after three consecutive errors. Split-half reliabilities were established as r_{sh} = .94 (addition) and r_{sh} = .92 (subtraction).

Results

Means, standard deviations, and correlations for and among all measures are shown in Tables 1 and 2.

Latent change score model

To examine the mixture of cognitive abilities and number abilities underlying young children's early mental arithmetic development and performance, a latent change score (LCS) model (McArdle & Hamagami, 2001) was computed (Fig. 1) using the Mplus 8 program with the maximum likelihood estimator with robust standard errors (Muthén & Muthén, 1998–2017). Maximum likelihood robust estimation was used to handle the missing data, which was low at most at 4.1% for the mental arithmetic measures during the third measurement point (Grade 2). LCS modeling allows for studying lon-

Table 1
Descriptive statistics for all abilities assessed in the study.

Ability	*M*	*SD*	Skewness (*SE*)	Kurtosis (*SE*)	Range
Counting knowledge	13.96	4.81	−0.449 (0.150)	−0.700 (0.298)	0–20
Symbolic magnitude comparison[a]	73.27	33.09	1.960 (0.150)	4.360 (0.298)	31–212
Number line estimation[b]	15.69	7.79	0.783 (0.150)	0.953 (0.298)	2.50–48
Nonverbal logical reasoning	12.62	3.89	0.732 (0.150)	0.744 (0.298)	3–27
Verbal working memory	7.11	3.70	0.377 (0.150)	0.291 (0.298)	2–19
Phonological processing	4.74	3.10	0.552 (0.150)	−0.454 (0.298)	1–13
Spatial processing	7.96	3.21	0.074 (0.150)	−0.553 (0.298)	1–16
General processing speed[a]	9.96	4.20	2.192 (0.150)	7.100 (0.298)	4–33
Mental addition in first grade	11.06	4.96	0.987 (0.150)	0.542 (0.298)	1–25
Mental subtraction in first grade	8.15	2.96	1.125 (0.150)	1.147 (0.298)	2–18
Mental addition in second grade	17.58	5.49	−0.294 (0.150)	−1.036 (0.298)	5–28
Mental subtraction in second grade	13.95	4.83	0.188 (0.150)	−0.628 (0.298)	5–25

[a] Time measure in seconds.
[b] Percentage absolute error.

Table 2
Correlations among the abilities assessed in the study.

Ability	2	3	4	5	6	7	8	9	10	11	12
1. Counting knowledge	−.60	−.32	.24	.24	.44	.26	−.30	.49	.39	.46	.43
2. Number line estimation[a]		.16	−.26	−.36	−.40	−.27	.28	−.49	−.42	−.49	−.44
3. Symbolic magnitude comparison[b]			−.26	−.07	−.13	−.19	.34	−.27	−.22	−.23	−.19
4. Nonverbal logical reasoning				.14	.26	.02	−.26	.31	.25	.30	.21
5. Verbal working memory					.34	.15	−.17	.22	.18	.30	.25
6. Phonological processing						.17	−.22	.39	.40	.32	.39
7. Spatial processing							−.17	.29	.26	.27	.22
8. General processing speed								−.30	−.33	−.24	−.25
9. Mental addition in Grade 1									.68	.57	.64
10. Mental subtraction in Grade 1										.49	.57
11. Mental addition in Grade 2											.65
12. Mental subtraction in Grade 2											

Note. N = 265. Correlation coefficients larger than r = .12 are significant at p < .05.
[a] Percentage absolute error.
[b] Time measure in seconds.

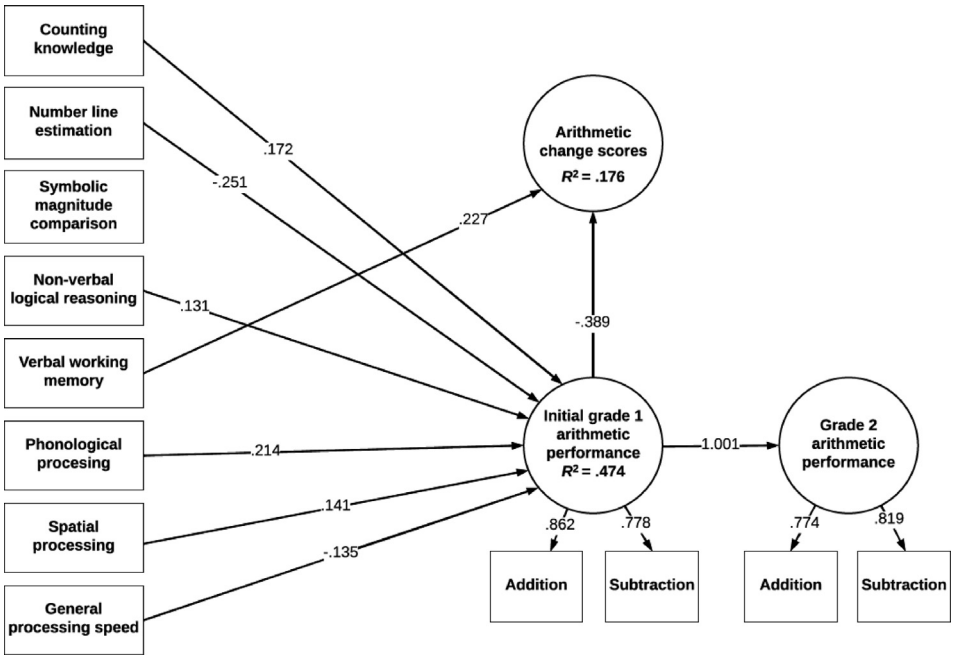

Fig. 1. The conditional latent change score model for mental arithmetic. Only significant (p < .05) associations (standardized estimates) are displayed. Errors are not shown for the purposes of description.

gitudinal change in ability, capturing this change as a latent factor, and determining the extent to which both initial task-specific performance and other predictors affect change over time. Key advantages include enabling longitudinal analyses between only two measurement points, the ability to determine both the variance in and rate of change, as well as describing the relationship between initial performance and the change score (including the ability to determine whether predictors at the first time point affect growth in achievement at the second time point) while minimizing measurement error (Kievit et al., 2018).

As a first step of analysis, a two-factor confirmatory factor analysis (CFA) was computed to establish if the mental addition and subtraction tests in Grades 1 and 2 could be used as indicators to attain reliable latent factors of mental arithmetic performance. Having highly reliable measures of arithmetic performance at these two measurement points is essential to obtain reliable latent change scores when performing an LCS model. The two-factor CFA displayed sufficiently high loadings on their latent factors (>.75, *ps* < .001) and a good overall model fit, $\chi^2(1, N = 265) = 0.047$, $p = .828$; comparative fit index (CFI) = 1.000; Tucker–Lewis index (TLI) = 1.012; root mean square error of approximation (RMSEA) = 0.000, 90% confidence interval (CI) [0.000, 0.099]; standardized root mean square residual (SRMR) = 0.001.

Next, an LCS model was computed. The model included eight predictors assessing counting knowledge, symbolic number magnitude processing, number line estimation, logical reasoning, spatial processing, verbal working memory, phonological processing, and general processing speed. The LCS model displayed a good overall model fit across all indices after defining the LCS and including the predictors, $\chi^2(17, N = 265) = 23.501$, $p = .134$; CFI = 0.988; TLI = 0.974; RMSEA = 0.040, 90% CI [0.000, 0.075]; SRMR = 0.023.

The LCS model accounted for 47% ($R^2 = .474$, $SE = .051$, $p < .001$) of the variation in latent mental arithmetic performance. Counting knowledge ($\beta = .172$, $p = .008$), number line estimation ($\beta = -.251$, $p < .001$), logical reasoning ($\beta = .131$, $p = .026$), spatial processing ($\beta = .141$, $p = .009$), phonological processing ($\beta = .214$, $p = .001$), and general processing speed ($\beta = .135$, $p = .010$) emerged as significant predictors of mental arithmetic performance in first grade. See Table 3.

The LCS model accounted for 18% ($R^2 = .176$, $SE = .086$, $p = .041$) of the variation in latent change score, and only verbal working memory ($\beta = .227$, $p = .031$) accounted for unique variance in latent mental arithmetic development (i.e., latent change scores) between first grade and second grade.

To control for the possibility of regression toward the mean, where initial performance may bias observed change scores, the approach by Sorjonen and colleagues (2022) was replicated. Two analyses were performed to determine whether observed associations between latent variables and predictor measures were unlikely to be spurious or caused by regression toward the mean. For Analysis 1, a positive association between significant predictors (here, verbal working memory) and change score should persist both when regressing change scores on initial performance and when the two factors covary (i.e., change scores are not adjusted for initial latent performance). For Analysis 2, the association between verbal working memory and change score should remain but flip to a negative sign when the causal direction is reversed, meaning that the model describes change scores from the second measurement point to the first one. Verbal working memory remained as a significant predictor of change score (with a negative sign for Analysis 2, as expected) across both analyses, suggesting a robust outcome.

Table 3
Standardized regression coefficients for each predictor across initial latent scores and latent change scores.

	Initial latent mental arithmetic scores in Grade 1			Latent mental arithmetic change scores		
Predictor	*β*	*SE*	*P*	*β*	*SE*	*p*
Counting knowledge	.172	.065	.008	.170	.149	.253
Number line estimation	−.251	.067	<.001	−.079	.145	.587
Symbolic magnitude comparison	−.076	.060	.207	.077	.138	.576
Nonverbal logical reasoning	.131	.059	.026	.040	.114	.725
Verbal working memory	−.046	.053	.386	.227	.105	.031
Phonological processing	.214	.066	.001	−.022	.117	.848
Spatial processing	.141	.054	.009	.021	.119	.857
General processing speed	−.135	.053	.010	.113	.108	.295

Note. pr^2, squared part correlations representing the unique contribution for each variable.

Discussion

The purpose of the study was to expand our knowledge concerning the mixture of cognitive abilities and number abilities underlying children's early mental arithmetic performance and learning by testing and comparing the number foundations account and the mixed foundations account. The results are discussed in relation to two theoretical accounts.

The relative unique contribution of number abilities and general cognitive abilities to early arithmetic performance and development

The LCS model demonstrated that six of eight abilities—counting knowledge, number line estimation, logical reasoning, spatial processing, phonological processing, and general processing speed—accounted for unique variance in mental arithmetic performance in first grade. This result is consistent with prior research and provides support to the claim, made by the number foundations account (Fuchs et al., 2010b) and the mixed foundations account (Geary et al., 2017), that arithmetic performance is supported by a mix of both number abilities and general cognitive abilities (e.g., Geary et al., 2017; Martin et al., 2014; Östergren & Träff, 2013; Xenidou-Dervou et al., 2018).

The LCS model revealed that individual differences in mental arithmetic learning (i.e., latent change scores) between first grade and second grade were supported by only one ability, verbal working memory, accounting for 5.15% (i.e., β^2 = .0515) of the variation in the latent change scores. This result aligns with Xenidou-Dervou et al.'s (2018) and Träff et al.'s (2023) findings, but it contradicts four prior developmental studies (Aunola et al., 2004; Fuchs et al., 2010b; Geary, 2011; Zhang et al., 2014) as well as the claim (Hypothesis 1) made by the two accounts that early arithmetic skill development is supported by a combination of both number abilities and cognitive abilities (Fuchs et al., 2010b; Geary et al., 2017; Lee & Bull, 2016).

The current patterns of predictor abilities align with the mixed foundations account (Hypothesis 2b; Geary et al., 2017; Lee & Bull, 2016) given that mental arithmetic performance was equally reliant on number abilities (9.32% total unique variance) and general cognitive abilities (10.11%), whereas mental arithmetic development was only supported by verbal working memory. These findings contradict the number foundations account (Hypothesis 2a; Fuchs et al., 2010b) and most prior studies (e.g., Aunola et al., 2004; Geary et al., 2017; Zhang et al., 2014) given that they highlight the key roles general cognitive abilities play in young children's mental arithmetic performance and learning that may be due to their low mathematical knowledge and skill levels (Geary et al., 2017).

The role of individual predictors of mental arithmetic performance and development

Consistent with von Aster and Shalev's (2007) model (Step 2) and prior research, counting knowledge accounted for 2.96% (i.e., β^2 = .0296) unique variance in mental arithmetic performance (Aunola et al., 2004; Geary, 2011; Xenidou-Dervou et al., 2018; Zhang et al., 2014). The importance of possessing sufficient counting knowledge skills is reasonable, assuming that first graders employ counting strategies (i.e., finger and verbal) to solve arithmetic problems (Geary & Hoard, 2005; Siegler & Shrager, 1984; von Aster & Shalev, 2007). The number line estimation task accounted for the largest amount of unique variance, 6.3% (i.e., β^2 = .0630), in arithmetic performance (cf. Fuchs et al., 2010a; Geary, 2011; Geary et al., 2017). This finding aligns with the fourth step of von Aster and Shalev's (2007) model, which states that children develop a spatially oriented symbolic mental number line that represents an understanding of the two core aspects of numbers: cardinality and ordinality. Given this, it is assumed that a high-quality ordinal (and cardinal) symbolic mental number line is important given that children are presumed to move mentally along it during arithmetic thinking (von Aster & Shalev, 2007; see also Lyons et al., 2014).

Similar to prior research, phonological processing emerged as a significant predictor of mental arithmetic performance, accounting for 4.58% (i.e., β^2 = .0458) of the variance (cf. Fuchs et al., 2010b; Martin et al., 2014). Theoretically, this specific aspect of language ability may have supported younger children's mental arithmetic performance by providing the distinct and accurate

verbal–phonological number representations necessary for efficient arithmetic problem solving (De Smedt et al., 2010; von Aster & Shalev, 2007).

Consistent with previous studies using mental rotation as an indicator of spatial processing, the mental rotation task accounted for 1.99% (i.e., β^2 = .0199) unique variance in mental arithmetic performance (Träff et al., 2023; Zhang et al., 2014). This connection may reflect that when young children solve arithmetic problems (especially multi-digit problems), they may transform the problem into spatial representations. This spatial representation is then matched against the spatially oriented mental number line (Step 4; von Aster & Shalev, 2007) to enable effective cardinal and ordinal number processing (cf. Barrouillet et al., 2008; Mathieu et al., 2016).

In accordance with prior studies, general processing speed emerged as a significant predictor of mental arithmetic performance, accounting for 1.82% variance (i.e., β^2 = .0182) (cf. Bull & Johnston, 1997; Geary, 2011). This finding is theoretically expected, supposing that nearly all academic learning entails rapid identification, activation, and access to information (Carroll, 1993; Geary, 2011; Gottfredson, 1997). In view of arithmetic performance, high general processing speed is required to enable rapid identification of number symbols (digits and number words) and operation symbols (+ and −) as well as activation of their underlying meanings (Bull & Johnston, 1997).

As hypothesized, logical reasoning ability accounted for 1.72% (i.e., β^2 = .0172) unique variance in mental arithmetic performance (cf. Geary, 2011; Jordan et al., 2010; Träff et al., 2023; Xenidou-Dervou et al., 2018). This relationship was expected given that learning and performing arithmetic should, in theory, impose several challenges to children's logical reasoning ability (Geary, 2013; Geary et al., 2017; Lee & Bull, 2016). To succeed in arithmetic, a child must understand the underlying logic of adding and subtracting, which requires that the child has a conceptual understanding of several key concepts (i.e., cardinality, ordinality, place value, the base-10 number system, and arithmetic operators). The child must also understand how different arithmetic operations are related (i.e., calculation principles; see Baroody & Dowker, 2003; Geary, 2013).

The finding that verbal working memory capacity of 5.15% (β^2 = .0515) supports mental arithmetic development is reasonable assuming that the working memory system provides a flexible and efficient mental learning workspace where new conceptual and procedural knowledge can be integrated with existing knowledge (cf. Fuchs et al., 2010b; Gathercole & Alloway, 2004; Geary, 2011). As such, this learning workspace may facilitate the child's ability to solve progressively more difficult arithmetic problems, by learning new effective strategies, selecting the most effective strategy given the task at hand, and shifting between strategies, as well as to execute them in increasingly effective ways (Baddeley, 1997; Geary, 2013).

Arithmetic performance versus arithmetic development

In keeping with prior developmental studies, especially Geary (2011), Träff et al. (2023), and Xenidou-Dervou et al. (2018), the results show that the mixture of abilities underlying mental arithmetic development and performance are rather different. A crucial finding is that neither of the abilities that accounted for variance in mental arithmetic performance accounted for variance in mental arithmetic development and vice versa. Therefore, a much smaller amount of variance in mental arithmetic development (18%) was accounted for compared with performance (47%) (cf. Aunola et al., 2004; Geary, 2011; Xenidou-Dervou et al., 2018; Zhang et al., 2014). One plausible explanation as to why this occurred is that the current study used a narrow arithmetic outcome measure to assess mental arithmetic at both measurement points, although the problems become progressively more difficult. Given the similar content of the arithmetic outcome measure at the first two time points, it appears that the more difficult problems the children encountered and managed to solve during the second time point did not substantially increase the demands on the children's number abilities and cognitive abilities except for verbal working memory. The fact that growth in mental arithmetic was only uniquely supported by verbal working memory may to some extent depend on the nature of the arithmetic task, that is, solving multi-digit problems through mental calculation. In relation to this finding, it should be noted that Fuchs et al. (2010b) previously observed that verbal working memory capacity contributes to 6-year-olds' development of double-digit mental calculation as well as development in arithmetic word problem solving. Verbal working memory has also been found to support

children's mathematical development (numerical operations test) from kindergarten to Grade 5 (Geary, 2011).

Limitations

It should be acknowledged that, compared with most prior research, the current study used a narrow mental arithmetic outcome measure to track children's development, which may reduce the possibility of generalizing the current findings to other aspects of arithmetic (e.g., complex written multi-digit calculation, arithmetic word problem solving). Thus, the mixture of abilities supporting other aspects of arithmetic might be different compared with those supporting mental arithmetic outcome. However, the current pattern of abilities supporting mental arithmetic performance is overall quite similar to research focusing on other aspects of arithmetic performance.

It should also be noted that several potentially key abilities (e.g., subitizing, visual–spatial working memory, shifting executive function) of arithmetic performance were not included in the current set of predictors. It is possible that these non-included abilities would contribute to growth in mental arithmetic development as well as performance.

A final note should be made concerning the nonsignificant contribution from the symbolic magnitude comparison task to mental arithmetic. This nonsignificant link may be partly due to the fact that response time was used as the outcome measure instead of accuracy. That is, a differently designed symbolic magnitude comparison task using accuracy as the outcome measure may have detected a contribution to mental arithmetic.

Conclusions

The current findings are theoretically important in relation to the cognitive resources underlying young children's early mental arithmetic performance and development. Mental arithmetic performance in first grade was uniquely supported by two number processing abilities (counting knowledge and number line estimation) and four general cognitive abilities (phonological processing, spatial processing, general processing speed, and logical reasoning). In contrast, mental arithmetic development between first grade and second grade was only uniquely supported by verbal working memory. These results demonstrate that the mixture of abilities underlying arithmetic development and arithmetic performance are rather different. Mental arithmetic performance in Grade 1 is equally reliant on a combination of both number abilities and cognitive abilities, whereas mental arithmetic development is not.

An important task for future research is to explain a larger amount of the variation in children's mathematical learning and development. To accomplish this, future research should, if possible, try to include multiple measures for each concept to allow for analysis on a latent variables level. Furthermore, it might be necessary to broaden the scope of processes above and beyond domain-specific number abilities and domain-general cognitive abilities. For example, it might be prudent to consider children's understanding of the equal sign, arithmetic principles, and laws as well as educational (teaching), affective, and social factors to account for a larger amount of variance in mathematical learning. This future work hopefully would lay the foundation for a new and expanded framework of early mathematical development in young children.

CRediT authorship contribution statement

Ulf Träff: Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Rickard Östergren:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Conceptualization. **Kenny Skagerlund:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Mikael Skagenholt:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Data availability

The first author has full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. The data are available upon request to the corresponding author by e-mail.

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